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Tungsten Alloy Electroplating Technology and Its Application on Downhole Tubulars

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Abstract

The maturation of oilfields often leads to increased water cuts or long-term exposure to corrosive media, intensifying downhole corrosion. This necessitates the replacement of standard carbon steel tubulars with expensive chromium or nickel alloys, driving up operational costs. Consequently, novel cost-effective corrosion control technologies are urgently needed. Tungsten alloy electroplating technology (TAET), a surface treatment method proven to enhance corrosion and wear resistance in other industries, is investigated for downhole tubulars' feasibility. Its mechanism involves forming a dense protective film upon initial corrosion exposure, inhibiting corrosive media penetration. The tungsten alloy plating has four main performance advantages including excellent corrosion resistance, great wear resistance, pretty smooth & dense surface and excellent adhesion. Taking a rolling exploration and development well ZSY-705 in the ZSY gas field as an application case, whose hydrogen sulfide (H₂S) content is 8.6%, comprehensive laboratory tests simulated downhole severely-corrosive conditions. Results demonstrated the tungsten alloy plated tubular (TAPT) samples consistently exhibited corrosion rates below 0.025 mm/yr., with only minor surface corrosion. Comparative tests showed its corrosion rate (0.0113 mm/yr.) significantly outperformed 13Cr (0.1599 mm/yr.) and S13Cr (0.0229 mm/yr.) samples. Pilot application involved deploying tungsten alloy plated tubing throughout the Well ZSY-705. To date, this installation has operated safely for over two years while achieving a 70% reduction in tubular costs compared to nickel-based alternatives. Laboratory and field evidence conclusively demonstrates TAET is a technically viable and economically superior solution for ensuring tubular integrity to a certain extent in complex downhole environments containing high levels of hydrogen sulfide, carbon dioxide, and bacteria.

Keywords: Tungsten Alloy, Electroplating Technology, Downhole Tubulars, Corrosion Prevention

Introduction

As numerous oilfields mature and produce more water, or are exposed to corrosive media, a growing number of wells experience severe corrosion, making corrosion control increasingly challenging^[1-5]. For downhole tubing and casings, the commonly used carbon steels are always substituted by chromium-based or nickel-based materials to enhance corrosion resistance, which highly increase the total operation investment^[6-8]. Therefore, it is necessary and valuable to find some novel tubular technologies in order to reduce well cost beneficially.

As a revolutionary achievement in surface treatment industry, the TAET has been successfully applied in fields such as engineering machinery and chemical equipment to improve the anti-corrosion and wear-resistant performance of the products, which provides us a supportive opportunity to solve downhole corrosion problems. This technology can effectively prevent the corrosive medium from penetrating into the base-body by forming a dense corrosion product film in the early stage of corrosion.

In this study, the technical principle and applications of TAET are overviewed and introduced firstly, then four main performance advantages of tungsten alloy plating (TAP) including excellent corrosion resistance, great wear resistance, pretty smooth and dense surface and excellent adhesion are summarized and analyzed one by one. Taking a rolling exploration and development well ZSY-705 in the ZSY gas field as an application case, whose hydrogen sulfide (H_2S) content is 8.6%, six groups of lab tests have been performed under different severely-corrosive downhole conditions. Test results showed that the corrosion rates of TAPT samples are all less than 0.025 mm/yr., and only slight corrosion occurs on the surface of samples. Three groups of lab tests for different anti-corrosion tubulars showed that corrosion rate of TAPT sample (0.0113 mm/yr.) is much lower than that of commonly used 13Cr (0.1599 mm/yr.) and S13Cr (0.0229 mm/yr.) samples. Subsequently, the tungsten alloy plated tubing has been favorably deployed in the whole section of Well ZSY-705. It has been operating safely for more than two years till now and the total tubular cost has been saved 70% compared with the nickel-based tubing, which perfectly showed the cost-effectiveness of this technology.

Technology Overview

Technical Principle

TAET is based on the principle of metal induced co-deposition. Elementary substance W (tungsten) cannot be electrodeposited from aqueous solutions, but can have co-deposition with elements such as Fe (Ferrum), Ni (Nickel), and Co (Cobalt). Tungsten (& nickel) alloy plating with both anti-corrosion and wear-resistant properties is prepared by reacting metal tungsten with nickel.

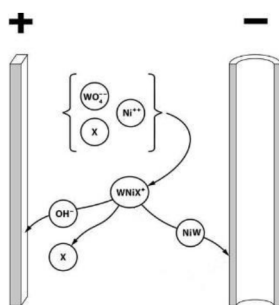


Figure 1—Schematic diagram of $Ni-W$ alloy electroplating principle.

The TAP exhibits a nano-crystalline structure characterized by high density and smooth surface morphology. During the initial corrosion phase, it forms a uniform corrosion product film that effectively

blocks corrosive media penetration through the plating into the substrate, so as to effectively protect the base-body and achieve the effect of anti-corrosion.

TAP is mainly composed of *Ni* and *W* elements, with some *Fe* and a small amount of trace elements such as *P* (Phosphorus) and *C* (Carbon). The content of tungsten is an important factor affecting the performance of plating. The higher the content of tungsten, the better the high-temperature oxidation and corrosion resistance. Due to high chemical stability, tungsten does not react with sulfur and can effectively inhibit corrosion in acidic media.

The composition of TAP is similar to that of common nickel-based alloys, and it has similar anti-corrosion properties, but pretty lower cost. If this technology is deployed on downhole tubulars for corrosion protection, the mechanism can provide superior anti-corrosion protection while simultaneously reducing friction and drag resistance, thereby significantly extending the service life in extreme environments featuring high temperature/pressure conditions and high concentrations of corrosive media economically.

Technical Applications

Tungsten alloy layer can be electrodeposited on the surfaces of ordinary carbon steel, low-alloy steel, aluminum alloy, copper alloy and other materials to improve anti-corrosion and wear-resistant performance of the products, especially to extend its service life in highly corrosive environments such as high temperature and high concentration of H_2S , Cl^- , and CO_2 .

TAET has been successfully applied in fields such as construction machinery, chemical equipment, marine engineering, military products and household items, achieving significant benefits in corrosion prevention and wear resistance.

As a revolutionary achievement in surface treatment industry, this technology also has been implemented on petroleum facilities and products for better anti-corrosion or wear resistance performance, such as pump barrels, downhole tools and valves (Fig. 2).



Figure 2—Tungsten alloy plated petroleum facilities and products: (a) Pump barrels; (b) Downhole Tools; (c) Valves.

Performance Advantages of TAP

The TAP combines advantages of metal tungsten and nickel, and has excellent properties such as high hardness, good high-temperature stability, and chemical stability. Combining high strength and high stability, TAET is potentially the preferred solution for anti-corrosion and wear-resistant treatment.

Excellent Corrosion Resistance

The TAP features a minimum thickness of 30 μm with excellent uniformity, delivering sustained high-temperature resistance up to 600°C for diverse thermal operating conditions. TAP is suitable for various highly corrosive environments, especially in chloride, sulfide, and carbonate media. Compared with other plating materials, the tungsten alloy has more significant corrosion resistance advantages.

Taking the actual data collected in PG gas field as an example, the TAPT samples were compared with the unplated tubular (UT) samples (carbon steel) through corrosion tests. As Fig. 3 shown, it was found that the morphology of TAPT samples and the thickness of TAPs almost have no changes. Experimental results demonstrate that the corrosion resistance (corrosion rate) of TAP in highly corrosive media exceeds that of conventional materials by over 100 times, matching the performance of high-grade nickel-based alloys.

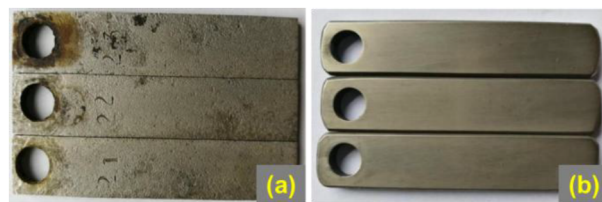


Figure 3—Post-test photographs of different samples: (a) UT samples; (b) TAPT samples.

Great Wear Resistance

The TAP achieves high hardness levels, with a minimum Vickers hardness of 600 HV that can reach up to 1138 HV through specialized processing treatments. Compared with other traditional metal plating products, TAP has obviously lower average friction coefficient and erosion loss.

Laboratory wear simulation tests were conducted to compare the dynamic wear behavior of TAPT samples and UT samples (J55 carbon steel) of identical specifications. The test parameters included a 400 N load, 2 hours of dry friction, 200 rpm rotational speed, and a 40 mm counterpart diameter. Post-test photographs of different samples are shown in Fig. 4, with quantitative comparisons presented in Table 1. According to the test results of wear weight loss, the wear resistance of plated samples are 5.9 times than that of J55 carbon steel samples.

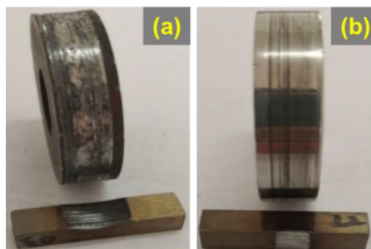


Figure 4—Post-test photographs of different samples: (a) UT samples; (b) TAPT samples.

Table 1—Wear simulation test results.

Test Samples	UT samples	TAPT samples
Wear Volume Loss (mm ³)	39.1	7.0
Wear Weight Loss (g)	0.296	0.050

Additionally, erosion testing was conducted to simulate downhole sand production conditions, employing a flow velocity of 15 ~ 20 m/s, sand concentration of 3.00×10^{-4} kg/s, and 20-hour exposure duration. The post-test measurements revealed that erosion marks of the TAPT sample are not obvious and its erosion resistance is 7.4 times than that of the UT sample according to test results of erosion loss.

Pretty Smooth and Dense Surface

The nano-crystalline structure, characterized by exceptional density and surface smoothness (Fig. 5), enables the formation of a uniform, compact corrosion scale during initial corrosion exposure. This protective barrier effectively impedes corrosive media penetration through the plating layer into the substrate, thereby providing robust substrate protection and delivering superior anti-corrosion performance.

Furthermore, the TAP exhibits significant tribological enhancements, demonstrating outstanding wear resistance with concurrent drag reduction capabilities, while simultaneously providing effective anti-scaling and paraffin-inhibition properties under wellbore production conditions.



Figure 5—Tungsten alloy plated product and samples: (a) Tubular product; (b) Material Samples.

Excellent Adhesion

The TAP and the base body are bonded by metallic bond, which has strong adhesion. The TAP is resistant to product body deformation, and both bending and tensile tests showed that the plating on the surface of the product is pretty difficult to fall off, as shown in Fig. 6.



Figure 6—Bending test of TAPT samples.

Additionally, hammering tests were conducted with Brinell hardness tester (load = 3000 N, indenter = 10 mm). The TAP at the indentations have no peeling or cracking changes, as shown in Fig. 7.

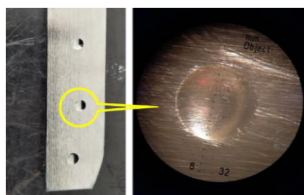


Figure 7—Hammering test of TAPT samples.

Application Case

Well ZSY-705, a rolling exploration and development well in the ZSY gas field, presents an exceptionally aggressive downhole environment with hydrogen sulfide (H_2S) concentration reaching 8.6%. This extreme corrosive condition necessitates superior corrosion resistance in completion strings. To ensure long-term production safety while optimizing completion costs, tungsten alloy plated tubing was selected for full-interval deployment. Implementation feasibility was validated through comprehensive corrosion immersion tests under simulated downhole conditions and comparative analyses of anti-corrosion performance across various protected tubular materials, culminating in successful pilot application throughout the wellbore.

Corrosion Tests Under Different Downhole Conditions

Per ASTM G111 guidelines^[9], corrosion immersion tests simulating ZSY gas field's representative service conditions were conducted under high temperature and high pressure (HTHP).

Different downhole conditions and the corresponding corrosion rates are listed in Table 2. Post-test corrosion rate characterization of TAPT samples confirmed mild corrosion behavior and the corrosion rates are all less than 0.025 mm/yr., only slight corrosion occurring on the surface of samples. The testing

results demonstrated suitability for sustained corrosion protection in complex operational environments characterized by elevated sulfide/CO₂ concentrations and microbiologically influenced corrosion (MIC).

Table 2—Corrosion tests under different downhole conditions.s

Simulated Downhole Condition	Temp. (°C)	Cl Conc. (g/L)	H ₂ S Partial Press. (MPa)	CO ₂ Partial Press. (MPa)	O ₂ Partial Press. (MPa)	Avg. Corrosion Rate (mm/a)	Evaluation Conclusion
1	200	150	3	10	-	0.0223	NACE RP 0775: Mild Corrosion (< 0.025 mm/a)
2	160	100	1.5	8	-	0.0191	
3	160	165	0.6	3.5	-	0.0065	
4	120	80	0.1	2	-	0.0044	
5	130	140	-	-	1.5	0.0047	
6	60	field-produced water (MIC)		0.5	-	0.0012	

Fig. 8 demonstrated that after corrosion immersion tests, the TAPT samples exhibited no observable blistering, cracking, or delamination. However, the surfaces displayed complete loss of metallic luster due to uniform oxide film coverage. Subsequent chemical cleaning revealed no detectable corrosion pits on the substrate surface.

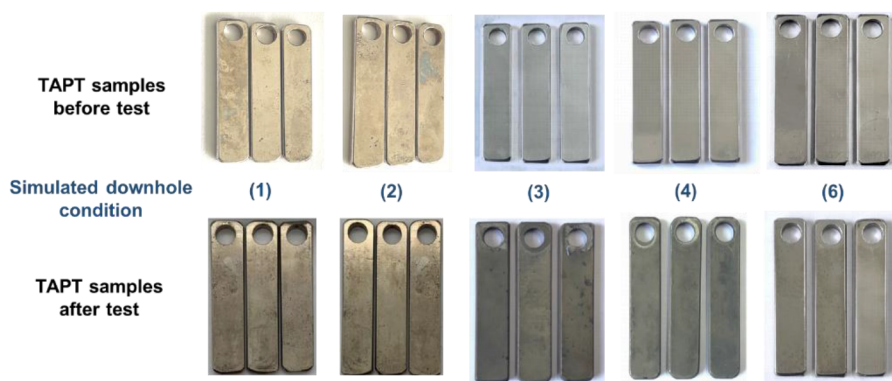


Figure 8—Comparisons of TAPT samples before and after corrosion immersion tests.

Corrosion Tests with Different Anti-Corrosion Tubulars

In downhole tubular applications, conventional carbon steel tubing (e.g., J55/N80 grades) requires substitution with corrosion resistant alloys (CRAs)^[10–11], typically chromium-based (13Cr/S13Cr) or nickel-based (Alloy 28/825) materials to withstand aggressive service environments. This material upgrade imposes significant economic impacts, elevating total capital expenditure compared to carbon steel alternatives while substantially increasing operational complexity.

To further prove technical feasibility of TAPT, corrosion comparison tests were carried out by selecting three commonly used anti-corrosion tubular materials. The simulated downhole conditions and testing results are listed in Table 3 and Table 4, respectively.

Table 3—Simulated downhole conditions for corrosion tests with different anti-corrosion tubular materials.

Temp. (°C)	Total Press. (MPa)	H ₂ S Partial Press. (MPa)	CO ₂ Partial Press. (MPa)	Cl Conc. (g/L)	Time (h)
160	10	0.3	5	25000	168

Table 4—Testing results for corrosion tests with different anti-corrosion tubular materials.

Anti-Corrosion Tubular Materials	Sample No.	General Corrosion Rate (mm/a)	Avg. Corrosion Rate (mm/a)
13Cr	01	0.2311	0.1599
	02	0.2128	
	03	0.0359	
	04	0.0257	
S13Cr	05	0.0256	0.0229
	06	0.0175	
	07	0.0023	
Alloy-28	08	0.0028	0.0072
	09	0.0166	
	10	0.0087	
TAPT	11	0.0138	0.0113
	12	0.0115	

According to [Table 4](#), it can be seen that the average corrosion rate of TAPT samples (0.0113 mm/yr.) is much lower than that of commonly used 13Cr (0.1599 mm/yr.) and S13Cr (0.0229 mm/yr.) samples, exhibiting superior corrosion resistance performance relative to both standard 13Cr and premium super 13Cr alloys under equivalent service conditions, which demonstrate that TAPT can be employed to the well ZSY-705 technically and beneficially.

Pilot Application in Well ZSY-705

Beyond corrosion testing and anti-corrosion performance evaluation, comprehensive assessments including mechanical property check, make-up and break-out testing, plating thickness optimization, and wear resistance analysis were conducted. This rigorous qualification process confirmed the suitability of tungsten alloy plated tubing with P110SS substrate, 88.9 mm OD, and TPCQ connections for Well ZSY-705 deployment. It demonstrated corrosion resistance equivalent to nickel-based alloys while maintaining critical thread sealing integrity and base material mechanical properties. Field installation protocols were established with specific quality controls and plating damage mitigation measures.

To ensure successful tubing running operations, implementation safeguards were executed through: rigorous risk identification, operational procedure optimization, plating thickness verification, controlled running speed, precise makeup torque determination, and sealability validation. These measures guaranteed operational safety and process reliability. The well-head running of tungsten alloy plated tubing for Well ZSY-705 is illustrated in [Fig. 9](#).

**Figure 9—Well-head running of tungsten alloy plated tubing for Well ZSY-705.**

Accordingly, the tungsten alloy plated tubing has been favorably deployed in the whole section of well ZSY-705. It has been operating safely for more than two years till now and the total tubular cost has been saved 70% compared with the nickel-based tubing, which perfectly showed the cost-effectiveness.

Conclusions

In this study, the tungsten alloy electroplating technology (TAET) is overviewed and analyzed in detail, both lab tests and pilot investigation on downhole tubulars have been conducted to demonstrate this novel technology is feasible and profitable for corrosion prevention. The following conclusions can be drawn:

- a. Utilization of TAET can electroplate a tungsten alloy layer on substrate surfaces, which may form a uniform corrosion product film that effectively blocks corrosive media penetration through the plating into the substrate, providing superior anti-corrosion protection.
- b. The TAP has four main performance advantages including excellent corrosion resistance, great wear resistance, pretty smooth and dense surface and excellent adhesion.
- c. Six groups of lab tests have been performed under different severely-corrosive downhole conditions and testing results showed that the corrosion rates of TAPT samples are all less than 0.025 mm/yr., and only slight corrosion occurs on the surface of samples.
- d. Three groups of lab tests for different anti-corrosion tubulars showed that corrosion rate of the TAPT sample (0.0113 mm/yr.) is much lower than that of the commonly used 13Cr (0.1599 mm/yr.) and S13Cr (0.0229 mm/yr.) samples.
- e. Pilot application involved deploying the tungsten alloy plated tubing in the whole section of Well ZSY-705, and it has been operating safely for more than two years till now and the total tubular cost has been saved 70% compared with the nickel-based tubing, which perfectly showed significant cost-effectiveness.

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